

QUAID-I-AZAM UNIVERSITY
DEPARTMENT OF COMPUTER SCIENCE (BS 5th)

Terminal Examination Fall, 2020

Course No. ST-101 Course Title: Probability and Statistics

Max. Marks: 50

Time Allowed: 3 Hours

Teacher's Name: Mr. Abdullah Iqbal

Date: 15-JAN-2021

Note: Please attempt five questions. All questions carry equal marks.

- Q.1 (a)** How could you define Binomial distribution as an approximation of Normal Distribution? **(02)**
(b) The international coffee organization reported that percentage people over age of 10 who drink decaffeinated coffee increased from 4% in 1960 to 15% 1985. Suppose in 1985 a random sample of 5 persons over 10 is asked if they drink decaffeinated coffee. What is the probability that at least two will answer yes? **(04)**
(c) A man goes by car to his office, and route through city center takes him, on average, 27 minutes with a standard deviation of 5 minutes. With the opening of a new ring road, the man can bypass the congestion of the city center, but journey now takes, on the average, 29 minutes with a standard deviation of 2 minutes. Determine which route is the better one if the man has **(i)** 28 minutes, and **(ii)** 32 minutes to reach his office for an appointment. **(04)**

- Q.2 (a)** Find the probability that the sum of the scores on three rolls of an ordinary fair die is less than 5. **(04)**
(b) Every Saturday a man invites his sister to the theatre or to the cinema. 70 % of his invitation are to the theatre and 90% of these are accepted. His sister rejects 40% of his invitation to the cinema. Find the probability that the brother's invitation is accepted on any particular Saturday. **(03)**
(c) After settlement, average daily wages in factory had increased from Rs. 120 to Rs. 180 and the S.D had increased from Rs. 2.5 to Rs. 3.75. After settlement the wages have become higher and more uniform. Comment. **(03)**

- Q.3 (a)** Differentiate between joint, marginal and conditional probability. **(02)**

- (b)** The following table shows the distribution of random variable X

x	1	2	3	4
f(x)	1/10	3/10	A	1/10

Find **(i)** the value of A. **(ii)** mean and variance of X. **(04)**

- (c)** Let X and Y have the joint probability functions given by

$$f(x) = \frac{xy}{66} \text{ for } x = 2,4,5 \text{ and } y = 1,2,3$$

Find marginal probability functions of X and Y, and find out if X and Y are independent. **(04)**

- Q.4 (a)** Describe a Poisson Distribution with the help of at least two real life examples? **(02)**

- (b)** According to national fire protection report more than one-third of all hotel fires are caused by guest room furnishing. Suppose that guest room furnishings caused a hotel fire is 35%. What is the probability that three of next four hotel fires will be caused by the guest room furnishings, assuming independence of events? **(04)**
(c) Given that the height of college boys is normally distributed with mean 5'-2" and standard deviation 4", and that the minimum height required for joining the N.C.C. is 5'-4"; find the percentage of boys who would be rejected on account of their height. **(04)**

- Q.5 (a)** What do you mean by the term probability? Explain with the help of at least two examples? **(02)**

- (b)** A bowl contains 16 chips, of which 6 red, 7 white, and 3 are blue. If four chips are taken at random and without replacement, find probability that: **(i)** each of the four chip is red. **(ii)** there is at least 1 chip of each color. **(04)**
(c) The probability that the lady of the house is home when the Avon representative calls is 0.6. Given the lady of the house is home, the probability that she makes a purchase is 0.4. Find the probability that the lady of the house is home and makes a purchase when the Avon representative calls? **(04)**

BEST OF LUCK